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April 10th, 2007

Radboud University Nijmegen: Geheugen (Memory)

Assignment 1

The 'second order conditioning' refers to a paradigm in which a subject make a certain association involving 2 stimuli and 1 response. At first a conditioned stimulus (CS1) is being paired with a unconditioned stimulus (US). When a connection between those two is made in the subject's cognition, the CS will lead to a conditioned response (CR) originally belonging to the US. (e.g. in Pavlov's experiment a dog starts to drool (CR) after hearing a bell (CS) because the association with food (US).). Next a second stimulus (CS2) is being introduced, being paired with CS1. After this is done in the right way, the exposure of the subject to CS2 will also cause a CR to occur. This effect could happen because of two reasons:

- CS2 triggers CS1, leading to a CR respond. This is called S-S)
- CS2 triggers CR directly, also leading to a CR respond even when CS1 is being extinguished. This is called (S-R)

A the description of both reasons reveal, extinguishing CS1 can be done to discriminate them.

Davey and Arulampalan performed a second order conditioning experiment between 3 different groups. Two groups were used to test wheiter the conditioning is S-S or S-R, and the third group, a so called control group, is used to show the second order conditioning effect at all. The control group participated in the experiment as normal as possible, but a main difference is being made in phase 1. In the all groups sessions are being held for every subject. A session consists of 10 trials that are seperated by empty intertrials of 30 seconds. In a trial a silde was shown for 6 seconds (CS1) followed by a very uncomfortable tone (US). However, for the control group this 2 events were chronologically randomized within a trial, making it very unlikely second order conditioning to occur, since the association isn't being made.

Another way to tackle this issue, involves a different approach in phase 1 for the control group. In this approach intially the condition of showing the CS1 directly followed by the US occurs, except for one major change. A random function that's equally distributed of 3 of the following options occurs:

- The CS1 and US are being exposed to the subject, just like the other groups
- Only the CS1 is being shown, not followed by a US
- Only the US is being exposed, without the CS1 occurring

The first of the 3 options, might give the impression that some second order conditioning might occur, but because of option 2 and 3, in which the CS1 and US are presented alone, this (potential) association is being extinguished. And even when it's not extinguished completely, it should still provide significant difference with the two other groups. One big advantage of this approach, compared to the currently used one, is that the conditions in which the control group operates are closer to the conditions of the two other groups. A (small) downside is the potential of the creation of an association being made.

Davey and Arulampalam assessed whether the response to the CS2 was based on S-S or S-R learning. In phase one an association between CS1-US is being made. In phase two one between CS2-CS1. In phase three the association between CS1-US is being removed. In phase 4 these steps were tested by exposing the subjects to CS2 and see if CR occurred. This was the case, which means that a S-R learning has happened, because S-S is out of the picture because of the removal of the CS1-US association. Another way to test this issue is to make a modification to phase 3. In this case the association between CS2-CS1 is being removed again (CS2 is shown without being followed by CS1), interrupting the CS2-CS1-US connection and forcing a relation between CS2 and US to not involve CS1.

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Assignment 2

The terms overshadowing[1], reduced overshadowing, forward blocking, and backward blocking can be explained by an experiment done by S. Vandorpe and J. de Houwer. For every phenomenon different trials are needed of 6 seconds each with intertrial time of 3 seconds.

In order to show overshadowing, at phase one pairs of images of foods (CS1 and CS2) are shown followed by a allergic or not test stimulus (US). Then in the test phase the conditioning of the CS1 and CS2 to the US is being tested individually and it turns out that often these don't produce the same strength. This is caused by a different saliancy, which means a person has subjective grounds at sensory level to attach higher value to one of the two CSes.

Second reduced overshadowing. As in the previous case, different items are being conditioned in pairs, except for one major difference. Prior to this phase a phase is introduced in which CS1 is shown individually, but without the US. It turns out that in the testing phase CS2 is conditioned more, because of the reduced effect on CS1 that's caused in the prior phase. This can be compared to a control condition of CS3 and CS4 in which just the paired phase takes place.

Third forward blocking. Conditions are similar to reduced overshadowing in the second phase and test phase, but a distiction is being made in the prior phase. First some pictures are shown (CS1) with their allergic property (US) and then the CS1 pictures are being paired with a new CS2 in the next phase, also with their outcome included. In the test phase it turns out that because of the prior phase including CS1, that CS2 has less priority and thus scores worse in conditioning. As in reduced overshadowing this could be check by introducing a control condition.

Finally backward blocking. In some way similar to forward blocking, except for the prior phase and second phase being switched. This means that CS1 and CS2 are shown with their outcome first, and in the next phase CS1 is shown with its outcome solely. This leads to worse results for conditioning of CS2 then skipping the second phase, because CS1 builds up more association.

As mentioned before, overshadowing gives different levels of conditioning to CSs when they are simultaneously paired to a US. This is caused by different saliency which means that different subjects have different ground on which they judge their sensory inputs. Using the Rescorla-Wagner model this can be explained by earlier associations of CS1 and CS2 build up by the subject each at different levels. This could be at any stage in the life of the subject and induces that it's a very subjective factor. For the theory part, this doesn't cause any problems, but for the testing and practical part it's something that needs to be taken in mind.

Although overshadowing doesn't lead to problems in the Rescorla-Wagner model, difficulties do occur in case of reduced overshadowing. When CS1 is shown initially without an outcome, the model doesn't change anything of the association between CS1 and US, since the initial value is zero already, and can't go any lower. This leads to the effect that the first phase of the experiment never happened in the model and that instead of reduced overshadowing, regular overshadowing is the outcome of the model's results.

There are different ways to tackle this issue and among them are the two following:

Other models have dealt with this problem by introducing non straightforward mechanisms in which CS1 becomes poorer in some kind of way when shown without a US. Then when CS1 and CS2 are paired, CS2 enjoys more priority

A second way to deal with it, which hasn't been named in the paper, is to instead of initial zero values of association to different connections, very small values to them. This makes since in some way, since the subject has experienced much in his life, and must somehow have some association between things, even when they are very loosely related.

1) <http://brembs.net/classical/overshadowing.html>

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Assignment 3

Concept acquisition is research that investigates how a subject (e.g. human or animal) learns how to group (often natural categories) different instances of objects. The different groups that are created in the subject's cognition are called concepts. Concepts are being created while the subject makes observations of different objects. The more objects a subject sees, the more advanced the performance will be on grouping these items. The learning of concepts can happen inductive (=without feedback and making hypotheses) or deductive(=based on facts).

Although associative learning theories have proven to be very successful within the concept acquisition area, some other theories and test results show different approaches could be made as well. Two downsides of associative learning theories include the obligation of learning being gradual and categories being formed in the cognition. These two issues are addressed in the following categories.

First all-or-none learning. Subjects are shown an artificial object and are told whether it's part of the category or not. Using this input, the subject makes up a hypothesis that matches the object belonging to the category or not. This is repeated and as long as the hypothesis is valid for new objects, no change is being made. Whenever the hypothesis isn't valid anymore, a new one is being made up. Depending on the complexity of the rule defining the category and the number of examples given and perhaps a coincidence factor, at one moment in time the stated hypothesis will match all members of the category. The cumulative results of many subjects learning this way, show gradual improvement, but looking at a single subject shows big fluctuation in performance, and suddenly when the right hypothesis is found a perfect score is being achieved.

And second the exemplar theory, which proposes that no categories are formed at all. Instead of that, a subject remembers a selection of past instances of various kinds and tries to match a new instance to a subset of the past instances that are similar (=group) by comparing its properties. This leads to a probability of the new instance belonging to the various groups, of which the highest value is most likely to be the corresponding group and thus category.

The following table is being used for a calculation with the Rescorla-Wagner model:

1. One large red triangle is in category A
2. Two small red triangles is in category A
3. One large blue circle is in category A

At every trial weights are being changed. Starting with trial 1:

$$\Delta V_{\text{one}} = 0,2 * (1 - 0) = 0,2 \Rightarrow V_{\text{one}} = 0,2$$

$$\Delta V_{\text{large}} = 0,2 * (1 - 0) = 0,2 \Rightarrow V_{\text{large}} = 0,2$$

$$\Delta V_{\text{red}} = 0,2 * (1 - 0) = 0,2 \Rightarrow V_{\text{red}} = 0,2$$

$$\Delta V_{\text{triangle}} = 0,2 * (1 - 0) = 0,2 \Rightarrow V_{\text{triangle}} = 0,2$$

$$\Delta V_{\text{tot}} = 0,2 * (1 - 0) = 0,20 \Rightarrow V_{\text{tot}} = 0,20$$

Next trial 2:

$$\Delta V_{\text{two}} = 0,2 * (1 - 0,4) = 0,12 \Rightarrow V_{\text{two}} = 0,12$$

$$\Delta V_{\text{small}} = 0,2 * (1 - 0,4) = 0,12 \Rightarrow V_{\text{small}} = 0,12$$

$$\Delta V_{\text{red}} = 0,2 * (1 - 0,4) = 0,12 \Rightarrow V_{\text{red}} = 0,32$$

$$\Delta V_{\text{triangle}} = 0,2 * (1 - 0,4) = 0,12 \Rightarrow V_{\text{triangle}} = 0,32$$

$$\Delta V_{\text{tot}} = 0,2 * (1 - 0,4) = 0,12 \Rightarrow V_{\text{tot}} = 0,32$$

Finally trial 3:

$$\Delta V_{\text{one}} = 0,2 * (1 - 0,4) = 0,12 \Rightarrow V_{\text{one}} = 0,32$$

$$\Delta V_{\text{large}} = 0,2 * (1 - 0,4) = 0,12 \Rightarrow V_{\text{large}} = 0,32$$

$$\Delta V_{\text{blue}} = 0,2 * (1 - 0) = 0,2 \Rightarrow V_{\text{blue}} = 0,2$$

$$\Delta V_{\text{circle}} = 0,2 * (1 - 0) = 0,2 \Rightarrow V_{\text{circle}} = 0,2$$

$$\Delta V_{\text{tot}} = 0,2 * (1 - 0,4) = 0,12 \Rightarrow V_{\text{tot}} = 0,44$$

Next when an object 'one small blue triangle' associative strength 'Category A' can be tested:
 $(0,32+0,12+0,2+0,2) * 0,44 = 0,3696$

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Assignment 4

A way to explain the process of making inferences about causes and effects can be done using an associative theory. When a cause is often followed by an effect, an association between the cause and effect is being made, such as the dog in Pavlov's experiment or a rat pulling a lever and being rewarded with food. Then when the same cause isn't followed by the effect several times, this association will become less and eventually virtually disappear.

Some objects against a purely associative account of causal inferences are the following:

- Using a purely associative account lacks of representing difference between causal and statistical relations.
- This approach makes it impossible to differentiate between causes and effect, while in a causal model relationships can have directions and thus be asymmetrical. This is tested in the semantic-memory task by Fenker et al (2005).
- Causal models can describe things that are useful to learn, such as different effects of one common cause, while knowing different causes of a common effect is not or less useful.
- Experiments focused on a difference between seeing and doing showed a difference in how relations are being stored after they've been done compared to just been observed.

Waldmann et al proved that animals can make causal inferences that are not based on simple associations. This was done in the following way: Rats went through an observational phase in which light was sometimes followed by a tone and other times by food. Then in the next phase when they heard a tone, they expected food. This shows they've learned link-by-link in the causal model ('seeing'). Next a lever was introduced in the cage, which caused the same tone when being pressed ('doing'). Pressing the lever made them less inclined to search for food, which means they thought they caused the tone and thus no food was expected. In a second study a causal chain tone-light-food was presented, and the rats expected food regardless of observing the tone or generating it with a lever. Comparing both situations shows that not just associations were being made, but also a way things are related such as described in causal network model, which means just associative information won't do.

As seen in the previous description of the experiment, it's not possible to just assume there's a bidirectional association between the light and the tone. This is the case, because as can be seen looking at the differences between the two studies, a difference connection is being made inside the rat between the tone and light. The first study shows that, even though a tone is presented after pressing a lever, the rats were less inclined to search for food, while if they were considered to be bidirectional, no matter when and how a tone is being generated, an expectation of food should occur, since the light is conditioned with the food.

The Anderson's books provides information about interactions between study (learning phase) and test. Different parameters have been found that dramatically influence test results in which scores were better when the study conditions were more similar to the test conditions. The parameters include among others: parameters of the environment (such as humidity) and parameters of the body state (drunkenness, fear, mood). In the paper of Vandersteenwegen (2006) acquisition and extinction of fear is being investigated with a focus on the role of context in both conditions being compared to the test conditions. In contrast of what Anderson's book provides, fear can be seen as a property or concept that's being learned (associated with some situation), and concepts effect included light/dark and no cue/cue being presented. Therefore the paper aims to reveal effects of context on changes in memory.

Renewal experiments investigate in what level after conditioning and extinction renewal effects can be detected. These experiments consists of 3 phases, namely conditioning, extinction and testing. Two different renewal methods are ABA renewal and ABB renewal. The letters ABA and ABB refer to the different phases of the experiment, in which A is a certain context and B a different context. The difference between ABA and ABB therefore occurs in the testing phase, in which in the former the testing phase context matches the conditioning phase and in the latter one it matches the extinction phase. Besides that one could note that in either way conditioning and extinction occurs in different contexts.

The experimental procedure used in the EC-group in the experiment by Vansteenwegen use two aspects that can be considered context, namely a preceeding cue and light conditions. When the light conditions are being considered either dark-light-dark or light-dark-light is that case which induces an ABA condition. But then on the other hand, when a focus is given to the cue, the EC group gets to deal with it in the 2nd and 3rd phase, making these equal, but different then the first phase. This implies a ABB renewal condition.

Three considerable and one purely theoretical reasons for the outcome of the experiment of Vansteenwegen include:

- Extinction occurred in a different context then acquisition and test. This might the retrieval cue to have a bigger effect when used in acquisition phase compared to extinction, since it matches the test conditions
- In the acquisition-cue group while looking at the associations between the CS and US, the cue can be seen as some sort of an CS that's never been extincted before and in the test phase, while in the other group everything being conditioned in phase 1 is extincted in phase 2. This might cause that in the acquisition-cue group the cue US relations might play a vital role in the performance above the Extinction-cue group.
- Due to the cue display in the extinction-cue group, the subjects might be 'confused' what the reason of the absence of the US is and associate this in a casual network with a cue. Then when in test phase the cue is shown again, less of an US expectation arises.
- Coincidence (Theoretically possible, but extremely highly unlikely). The distribution of the subjects over the groups would then be in a way in which almost all the subjects in one group would have very different properties for some reason (genetics, history) than most subjects in the other group.

N.B.: Any combination of all named above factors with each having a certain weight is possible, inducing virtually unlimited combinations of other distinctable reasons.

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Radboud University Nijmegen: Geheugen (Memory)

Assignment 6 (part 1)

In his book Anderson mentions the rise and fall of the theory of short term memory. This rise and fall refers to the proof of existence of short term memory, which has been empirically tested over time with contradicting results. Rundus(1971) showed that more often the numbers were rehearsed out loud the better they were recalled and the Atkins and Shiffrin theory from 1968 shows a separation between long term memory and short term memory, based for a big part on the recency effect. Contrary to that Glenberg(1977) showed in experiment that the amount of rehearsal had no influence on the storage when there's no complete focus on what to be studied. Craik and Watkins (1973) showed that there was no relationship between how long something has been in short term memory and how well it's stored in long term memory. And finally Neisser showed that even though he had read prayers at meals around 5000 times in 25 year, his performance was very poor on memorizing them. The sequence of these theories have caused a at first a believe (rise) in wheter short term memory exist and later a disbelieve (fall) in its existance.

According to Williams e.a.'s view and also the model of working memory developed by Baddeley, the working memory is conceptually defined as a system consisting of one central executive and a number of subsystems that are coordinated by the central executive. Two of those subsystems include the phonological loop and the visuospatial sketchpad. Relating the previous to the executive function theory, means that poor performance of individuals with autism on problem solving tasks (which are demanding on using working memory) are caused by malfuction on one or more of these subsystems, which has been doubted and impossible to yield evidence of in studies of Strayer(2001) and Russel et al (1996). Anderson explains the model of Baddely and the Sternberg Paradigm in his book to share different views on the concept working memory. A point in which Anderson agrees with Baddely is the existence of a central executive and subsystems to aid it in storing short term information. The information is stored by neural activation and patterns of activation can also cause changes in synaptic efficacy, which results in long term memories.

Williams mentions different views on disorders in WM and EF of which several contrast. Benneto et al (1996) and others showed in their studies that people having autism scored worse then controls on planning and problem solves tasks, which is thought to be caused by a deficient working memory. Ozonof et al. (1991) showed, handling the principles the EF theory, that this problem is caused by an inability to store relevant or correct information in the articulatory loop or the visuospatial sketchpad. In contrast to that Ozonof&Strayer (2001) and others couldn't find any evidence in their studies about the quality of working memory having to be the cause for the previously mentioned problem. To support this view, other studies have empirically shown that the lack of performance on the EF is actually caused by incapacibilities in conceptional reasoning and planning. These findings appeared more and more when complexity was increased. Recently inconsistent evidence have been found about autism. In a spatial memory span task (Ozonoff&Strayer 2001) including the recalling of the location of 3 till 5 geometric shapes on a computer screen, autisms' performance equaled the control group, while in the delayed oculomotor response task (Goldman-Rakic 1982) in which a subject has to remember a location and focus on it, autistic people scored significantly worse then the control group.

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Radboud University Nijmegen: Geheugen (Memory)

Assignment 6(part 2)

There are different models of working memory, but the most commonly used is the one designed by Alan Baddeley, initially consisting of 3 units namely: a central executive, visuo-spatial sketch pad and a phonological loop. In which the former one is connected in two ways to the 2 latter ones, also known as subsystems. The model is used because it is able to integrate many findings that have been done in the area of working memory of which the previously named findings. In the report of Williams et al. (2005) people with autism are being investigated and compared to a control group. This is done to detect whether autistic people show different results regarding the 2 previously named subsystems. When this is the case, the autistic subjects can be used for different tests, in order to see how they perform on several tasks compared to the control group. The difference in performance can then be contributed to the lack of functioning on the respective system. In the previously mentioned report of Williams he concludes that people with autism score equally well in tasks requiring the articulatory loop but lacked performance on tasks involving visuospatial sketchpad. Taken in account the just previously mentioned, it's still hard to define right tasks to deal with this findings, because the both subsystems can relieve each other with several tasks. For example spatial information can be stored in a verbal way and chronological time in the articulatory loop can be stored as space. This makes it very hard to measure performance, since it's virtually impossible to tell how the information is being processed during a defined task.

Focusing at the test data of the different experiments, it's worth to have a look at how these reflect to the previous paragraph. In the WMS Letter Number task and WRAML Number/Letter tasks, performance of the autism group basically matches the control group and thus no real distinction could be made. Reviewing both tests, it's very reasonable to assume that the articulatory loop is likely to be used often in these tasks and can for autistic people at least at this area considered to be functioning. This matches the conclusion of the authors. In contrast to that, two other tasks named WMS Spatial span and the WRAML Fingers offer significant difference of performance in autistic people and the control group, in which the former scored much worse than latter. Also being concluded the report.

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Radboud University Nijmegen: Geheugen (Memory)

Assignment 7 (part 1)

People who have integrative agnosie, suffer from a disorder at neurological level. In the article of Behrmann et al(2006) a person S.M. got involved in a motor vehicle accident, which caused brain damage to him. Consequences of integrative agnosie emerge in case of object recognition and especially when an object consists of more parts that are organised in a certain spatial way. Tasks involving this aspect reveal impaired behavior. The authors of the previously mentioned article want to address several questions regarding this recognition. Mainly they want to investigate whether a different mechanism and part of the brain is responsible for recognizing parts of an object versus determining the spatial arrangements of the different parts forming the object (aka binding local parts). Although this is possible to do, it's impossible to prevent this question leading to more questions. When difference in performance occurs, which actually happens in the experiment, what causes it? In order to answer this second question, some extra research was needed, because the lack of performance on the spatial tasks by S.M. could be caused by simply having a more difficult tasks (different parts to be recognized in an object versus just recognizing a single part), which means it could still have been done by same parts and functions of the brain. This issue has been solved by having different people with same criteria as the previously used control group doing an almost identical experiment. The difference is the exposure in test phase, which has been set to 100ms. This resulted into measuring the false-identification rate of the normal controls compared to S.M. and induced into knowing that the difficulty of the task had no influence on the result and thus was caused by separate systems for recognizing parts and recognizing the spatial arrangement of the parts that form the object.

The above mentioned refers to investigating the likelihood of two visions that contrast with each other. The first assumes that objects are represented as a collection of parts in which coding happens independently while another assumes that recognition of an object is done on the object as a whole, without having a need to distinguish separate parts.

Due to the impaired behavior of S.M., it becomes possible to notice performance differences in some tasks and equal performance in others compared to the control group. This occurrence makes it possible to separate different tasks as having different areas and functions in the brain. Knowing this, can lead to a model of recognition in the brain with different units for different kind of tasks.

In the test phase of the experiment in the previously mentioned article, 2 distinct phases can be labeled named the No-Icons test block and the Icons test block. In the former one every subject had to deal with 48 target trials and 48 distractor trials in random order. The purpose of this phase was to recognize the different target items and label them in the right category by pressing a corresponding key on the keyboard. Both accuracy and speed were considered important for measuring performance. The Icons test block was very similar to the previously mentioned, except for the studied views of the 4 target items being available on top of the screen for display all the time during the phase. The 48 target trials consisted of different combined properties: 4 different targets and 6 different angles of view, while the 48 distractors had different properties. 8 distractors were the so called PC distractors in which a part of a target object is being replaced by a part of another target and 8 RC distractors in which the parts of the target items were spatially being rearranged, but not replaced by different parts. These 16 (8+8) distractors were shown in 3 viewing angles, resulting in 48 distractors. The test phase has been performed by the previously mentioned S.M. and 42 controls. Due to the sufferance of S.M. from integrative agnosie different performance could be expected. Results indicate that in either phase S.M. scores significantly worse on the relation changed distractors compared to the part changed ones. Putting this result in the perspective of the central question, one may conclude that it's likely that objects are represented as a collection of parts whose relations are coded independently.

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Radboud University Nijmegen: Geheugen (Memory)

Assignment 7 (part 2)

In regard of memory, the findings tell us that a separation is made between discriminating spatial relationships and identifying specific arrangements of image features. This means there are different functions responsible for it and thus different representations are made in memory of both features at a different locations.

Finally the view(s) of the authors of the article can be compared to the book of Anderson. Anderson focusses on the difference between recall and recognition, in which there are more cues available to gain information from and thus leading to better performance in the latter one. Anderson doesn't consider any properties of words, such as a word consisting of letters and how it's coded in memory. This aspect could be ignored somehow, since the task of recognizing words can be considered relatively easy. In contrast to Anderson's view, the authors of the article focus on the encoding difficulty and the way things are encoded for recognition. The number of items to be recognized is small in this case (four), but the kind of objects (3 dimensional objects consisting of parts that are spatially arranged) might require more encoding than words. This results in the view that different systems are responsible for encoding. Both views don't contrast each other, although in cases in which the encoding task is hard, recall can even score better than recognition. This occurs in an experiment involving an icons test phase and a no-icons test phase in which S.M. who's diagnosed to have integrative agnosia scores better on no-icons (free recall) than icons (recognition).

In case of the retention of memories, Anderson mentions context as cause of interference. To clarify his meaning he uses an example in which subjects study two different lists of words. The first list (called A-B) may contain word combinations like frog-door, house-key as long as every second word is different then every first word ever used. After this studying the control group gets to take as rest, before attending the final test. The other group however, gets to study a second list of words (C-D) which does not contain any words being used in the first list and once again having every second word being different from any used first word. Finally the words of A-B are being tested, in which the subject needs to reply with the right B word to a given A word. It turns out the test group scored worse then the control group on the test, because of the retroactive interference of the C-D list, even though none of the words is being used in the test that's in it. This is where the context aspect shows up. The word lists have been studied in the same laboratory and while studying the words there, the laboratory can become an element that aids the recall of words in the test. The laboratory becomes a so called context cue. Other context cues are the subject's mood, hunger and pain. Because the words of the C-D list are studied in a context with many aspects being almost the same, the association needs to be shared between both lists and therefore is weaker, inducing worse results. Finally in Anderson's opinion the context is basically anything that reaches are senses while making associations between the words (=studying), resulting in these contextual items being involved in the association process.

In the papers of Bird et al and Cipolotti et al. 3 different people with neurological impairments have been tested on several areas. The following test results were recorded:

Name	RH	JV	VC
Words	Good	Poor	Poor
Scenes	Poor	Poor	
Faces	Okay	Poor	Okay
Buildings			Poor
Landscapes			Poor

In which good means matching the controls well, okay scoring slightly worse then the controls and poor scoring significantly worse then the control group.

The scores not matching the control group can be contributed to the following factors for the different people: RH suffered from selective right sided hippocampal damage, JC had bilateral hippocampal damage and VC selective damage in the hippocampus bilaterally.

Note that buildings and landscapes match scenes in some way, but for clarification the different terms that have been used among the different experiments have been maintained.

The various experiments among the above listed people and control groups were held to contribute different impairments to different parts of the brain. The findings appeared to reveal that the right hippocampus contributes to topographical performance, the left one to some verbal materials while face recognition may be operated in areas outside the hippocampus. Remarkably there seems to be a dissociation between topographical tasks and face recognition, both behavioral and anatomical.

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Radboud University Nijmegen: Geheugen (Memory)

Assignment 8 (part 2)

To support the analysis of these experiments ROC-analysis has been used. This term refers to a way of analysing in which 4 different possible outcomes per test entity could be differentiated. These are often plotted in 2 by 2 matrix, in which the columns represent the factual value of the experiment (the value the subject is supposed to show when it's trained well) and the rows the actual reply of the subject. In the left top and bottom right field of these matrix they match, and the higher these values are compared to the other 2 values, the better the performance. Using the values and knowing the right calculations the accuracy, specificity, positive prediction value, negative prediction value and others can be calculated.

Anderson states in his book that context is being stored in the association, both in recognition as in familiarity. Experiments have shown that when this happens in divided attention, sometimes only the familiarity is stored without its complete contextual source, leading to false implications. In the papers previously mentioned, a distinction is made between various categories, which are differently influenced by impairments of one or two of the hippocampi. This induces that Anderson's view might be a little too straightforward, since no anatomical dissociation is mentioned at all. This dissociation makes it likely to expect different cognitive ways in which the humans deal with information on different kinds of contexts and categories of information. Something that remains future research right now.

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Assignment 9 (part 1)

Several experiments have shown that specific conditions can lead to false memories. The first thing that can be concluded from this, is that the human memory is never 100% accurate and thus isn't worth to be trusted that way. From a cognitive perspective it means that somewhere or in more places at the encoding process, retention process and retrieval process mistakes are being made. Many scientists, of which Foley McDaniel and Pressley, contribute these mistakes to imaginal codes strategies, in which information to be encoded is run through some process of imagination (for example putting the concepts to be remembered in a story plot) to reduce the load of information that needs to be stored. Later experiments have shown that the encoding process often makes such inferences.

In his paper of 2006, Foley offers the readers 3 different experiments. In her experiments she aims to create different situations in which his subjects need to encode, retent and retrieve information and by using so called lures, she tries to test the accuracy of this process. The focus of the experiments is mainly on the encoding process, which varies from function only (describing a function for each object) to generic imagery (generate an image for each object) to self referential imagery (connecting the image in some way to your life) till a condition in which the descriptions are made by participant (descriptions by non subjects are given to support the encoding of an image process). These different situations led to different results in the relative amount of false memories. These experiments are useful for understanding our memory, because different situations lead to different results while as many other factors as possible (the objects to be encoded, the laboratory enviroment) remained constant. The cases that are most interesting are the cases in which the relative amount of false alarms was high, because apparently in these situations something is being done that's opposed to our nature. For example, in the self referential imagery condition of experiment 2 the number of false alarms was relatively high due subjects subconsciously making assumptions about the situation due own former experiences.

The previously named experiments of Foley made use of the DRM paradigm. The DRM paradigm provides subjects with a list of words, in which every word presented is an associate of a word that turns out to be critical later in the experiment. For example: bed, rest, dream, snooze, blanket, doze, slumber, snore, nap, awake, tired and peace of the list of source words and the critical word being sleep. After that the subject is being tested on have stored the words in the learning phase by including different lures to the original group of words and asking randomly items from this combined list. In the previosuly mentioned word list, subjects have a tendency to judge the word sleep as being studied in the learning phase, due to the many associations of the words studied to sleep. The word sleep is called the critical lure in this case. In Foley's specific case, Foley tested the source memory in a few different variables, namely: Proportion correct yes to presented items (items correctly recognized from learning phase), proportion incorrect yes to critical lures (as explained before), proportion incorrect yes to control lures and new instances. The term source in memory refers to what has actually been chosen to encode. In Foley's different experiments every different condition, such as function only and generic imagery, leads to a different source, because memory is triggered in a different way in every condition, even though the intended to be remembered concept is the same amoung the various conditions.

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Radboud University Nijmegen: Geheugen (Memory)

Assignment 9 (part 2)

The experiments provide extra information with regard to the false memory syndrome for the following reasons:

- The experiments (try to) create a false memory on small scale
- The experiments give insights in the encoding, retention and retrieval process and errors that might arise in every phase
- The experiments test different situations in which the false memory rates vary. This information could be used to predict someone's accuracy of memory being in a certain context and everyone experiencing something in a certain way

With regard to someone having false memories of abuse in their youth some similarities and differences can be drawn compared to the experiments of Foley. The similarities include:

- The subject and the person thinking to be abused might be affected by the way they are questioned about potential facts they have experienced in their past
- The DRM method could be used in both cases in order to test the reliability of the person's judgement
- The person expressing its memory, truly believes the fact mentioned has happened in the past

The differences are:

- The history of the thinking to be abused person normally involves a lot of emotional arousal, which results in a certain way of dealing with senses in cognition or as Anderson states it psychological distress
- The time span of the fact being remembered varies much in both cases, leading to potential differences in the retention function
- In cases of people with potential abuse, the true facts are often unknown, while in an experiment setting all history facts can be recorded and be used to monitor performance

Finally a comparison of Foley can be made with the book of Anderson, regarding false memories. Both Anderson and Foley are convinced that people can be made to remember things that never happened. In addition to that, they both believe that errors happen due a lack of performance in encoding, which is caused by not exactly being able to identify the source of the information given to the senses. Differences occur in opinion when the cause of this impairment in performance is being discussed. Anderson mentions something called the arousal state much for determining encoding quality, which is a mechanism of nature to judge the importance of information entering the senses. For example, people who have experienced an earthquake seem to remember many details, also known as flashbulb memories. On the other hand, Foley implies, also noticable from the way he set up his experiments, that much of the encoding part has to do with fitting the sensation into current concepts. The results of his first and second experiment show a comparison of self referential and generically created images for memorization leading to different results in every experiment. Foley suggests that factors causing the differences might be the encoding time and the amount of imagination used to encode the sensed concepts into the current knowledge in long term memory.

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Assignment 10 (part 1)

Vakil et al tested in his paper of 2004 how (partly) malfunctioning of the right and left basal ganglionic part of the brain impaired procedural learning. Three groups, a group with lesions at the right BG, a group with lesions at the left BG and a control group, participate in different tests to compare performance between the different groups. The paper deals with 3 test in order to investigate this, namely: The tower of Hanoi puzzle, mirror reading and Porteus mazes. The tower of Hanoi puzzle deals with a number of discs (in this case 4) of different sizes and 3 stacks, having the goal to move all the discs from a beginning stack to eventually a final stack with the restriction of bigger discs not allowed to be placed on smaller ones. Procedural performance is examined by measuring the number of moves and the time to solve this puzzle. The second tasks, mirror reading, involved reading words that were mirrored. The speed and error rate of the task were measured to evaluate procedural performance. This was done both with new words and old words, to see potential learning. Finally a Porteus mazes task was done by the subjects. This task is about solving a maze with dead ends alleys. Entering such an alley was considered an error and was being measure. Above that the time of solving the maze was being investigated as well. This measuring happened on different trials, to detect learning. The variety among the tasks makes it possible to contribute impairments on specific tasks by a certain group to properties in which that task differs from the other two tasks, making it possible not to just detect impairments on procedural learning but also to address them to a certain part of procedure needing to be done.

The reason why these specific people besides the control group are chosen is mainly based on previous research and statements made by other researchers. As the above paper states, various studies have implicated a dominant role for the BG for procedural learning, mainly based on results with animals, patients with Parkinson or Huntington disease. All the subjects not being in the control group, have been examined to have structural damage on either the left or the right part of this area and are thus, assuming the previously mentioned is true, likely to show impairments on the 3 tasks. In contrast of this argument those chose this group of people there is the fact that people suffering from Parkinson or Huntington disease may have damage that extends the BG region. Although this makes it unsure whether the impairments on procedural tasks depends on the BG region, it made it extra important to investigate the subjects used in the experiment, to obtain more security on this issue. Considering all the previously mentioned in this paragraph, it's been very important to find patients with left or right BG lesion and no absence of mass there, but with no neurologic and/or psychiatric history. This to make is most likely that all the patient's impairment can be contributed to either the left or right BG.

After the 3 tests are done, the results of them can be compared. In the mazes and reading test the left BG group generally scores better then the right BG group, while both being outperformed by the control group. An exception to these findings occurs in the testing maze results being part of the maze experiment. In this case current knowledge obtained in the training maze needs to be 'converted' to a completely new situation, namely the testing maze. This is very unlike the other experiments performed, in which repetition lead to improvement. In the tower of Hanoi experiment, unlike the other two, no real distinction between the performance level of the left and right BG group can be made. The left BG group starts off worse, but on succeeding trials performance seem to level with the right BG group. This could have two reasons: Certain properties of the task are unlike the other two tasks or the task is not completely appropriate for procedural learning. The author of the paper contributes the different performance to a distinction between two aspects of procedural task performance, namely an abstract and specific part. Using this view to explain the mirror reading, which can be seen as a specific tasks (mirroring letters is the same thing all the time) and the training maze (which doesn't change on any trial) the better performance of the left

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Radboud University Nijmegen: Geheugen (Memory)

Assignment 10 (part 2)

BG group makes sense. Switching back to the Hanoi task, how can the results be explained? In my view the subjects of all group start off with a very abstract task, in which a certain goal needs to be achieved within certain constraints, but without knowing heuristics or guidelines to achieve the goal state, which is considering the above clearly an advantage to the right BG group. After several trials the left BG group shows a strong learning effect. I believe it can be explained in terms of switching from an algorithm to a more close to discrete (specific) task. Every tower of Hanoi problem can be represented in a graph containing all possible states and by sufficient practice for every of this states the optimal move can be learned (policy). This eliminates any need for abstract thinking. In my opinion it would be interesting to do the same experiment again, but only instead of 4 discs use 6 or more. This will reduce the possibility of the previously mentioned graph being learned by the left BG group and might show impaired performance for many trials on this group.

The book of Anderson uses the terms implicit learning and skill learning. Anderson's view of skill learning can be applied to the Hanoi tower game. In his view there are 3 stages which subjects can go through in case of repetitive rehearsal of the problem, namely: cognitive, associative and autonomous. In the first stage a general approach is being used, that is not specific to the domain. In case of the Hanoi tower problem, deep reasoning and trial and error might lead to the conclusion that detecting subgoals might be a good strategy. In case of moving four discs from A to C, moving the top 3 discs to B might be a good start, since it's the only way to move the biggest disc to C at all. Another major property of this stage is to reduce the difference between the current and goal state, but using this heuristic too much might lead to problems. For example, consider the starting state of the 4 disc Hanoi game. The just mentioned approach makes it very legitimate to try to move disc 1 to 3 to location C, which turns out to be very useless. Next the second stage called the associative one. In this stage the verbalisation of what needs to be done disappears and some sort of state diagram with different connections is being mapped in the brain, which can somehow be compared to a maze. The different states don't have a name, but are connected to a state in the real world and somehow have an identification to achieve that and can be recognized directly. When the domain specific procedures are being learned, proceduralization occurs. Finally the autonomous stage. The amount of cognition slowly disappears and many things seem to happen automatically and are hard to stop (for example switching off the light after going to the toilet). Another important property is losing the ability to describe what is actually done (i.e. a chess player making a certain move). This is when implicit learning happens and implicit memory is being generated. Implicit memory consists of facts that can't always be retrieved directly, but of which subjects can somehow prove they have that knowledge available. Some examples include: typing keys blindly on the keyboard without being able to construct a keyboard map and riding a bike without exactly being able to describe how to not fall off. For the Hanoi game this phase would mean that a player would perhaps be able to only solve the puzzle when physically playing but no being able to write down a solution tree.